

utdf2gmns: A Python Package for Mobility Simulation from UTDF to SUMO

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Summary

UTDF2GMNS¹ implements an automated workflow for network coordination, traffic signal integration, and traffic flow conversion from Synchro to SUMO. The process begins with a comparative analysis of network topologies, data representations, and signal timing schemas in both environments. Converting Synchro UTDF data to a network ready for microsimulation poses several challenges, including accurate signal integration, spatial transformation, and preservation of turning flow fidelity. Signal conversion represents a primary bottleneck, as it demands precise mapping of phasing plans, timing parameters, and coordination strategies to ensure valid simulation results. Network conversion is further complicated by translating Synchro's relative coordinate system into georeferenced formats compatible with geographic information system tools. Furthermore, accurate transformation of turning movement data is essential for realistic intersection modeling.

Existing methods address only isolated conversion tasks and lack a fully automated, scalable end-to-end workflow. To fill this gap, we introduce [utdf2gmns](#), an open-source Python library that automates the transformation of Synchro UTDf files into GMNS-compliant SUMO networks. [utdf2gmns](#) provides automatic geocoding, Sigma-X engine integration for intersection analysis, robust SUMO network generation, and extensibility to other microsimulation platforms. Future work will extend support for adaptive signal control, integrate real-time data inputs, and enhance interoperability with additional simulation frameworks to promote reproducibility and collaborative traffic-modeling research.

Statement of need

Traffic microsimulation is essential for evaluating and improving urban transportation systems by providing high-resolution analysis of flow, congestion, and infrastructure performance. Such simulations depend on precise modeling of signal control, network geometry, and turning movements. Although Synchro's Universal Traffic Data Format (UTDF) delivers comprehensive intersection data, converting UTDF into simulation-ready networks remains manual, labor-intensive, and error-prone, limiting seamless interoperability with microsimulation platforms.

Several critical challenges remain when converting Synchro UTDF data into microsimulation-compatible networks, such as those required by Simulation of Urban Mobility (SUMO) (Lopez et al., 2018). First, accurate signal conversion demands detailed extraction and mapping of phasing, timing, and coordination parameters into standardized control formats; errors here can substantially degrade simulation fidelity. Second, network conversion requires transforming Synchro's relative coordinate system into georeferenced longitude–latitude coordinates for

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39 seamless GIS integration, a labor-intensive and error-prone process that limits scalability. Third,
40 realistic intersection dynamics hinge on precise turning movement conversion, which typically
41 involves extensive manual preprocessing; inaccuracies at this stage can propagate through the
42 simulation, undermining the validity of subsequent analyses.

43 To address these gaps (Ban et al., 2022; Coogan et al., 2021; Singh et al., 2017; Udomsilp et
44 al., 2017; Zhang et al., 2024), we present utdf2gmns (Luo and Zhou 2022), an open-source
45 Python tool that automates the conversion of Synchro UTDF files into GMNS-compliant
46 networks (Smith et al., 2020) and generates simulation-ready inputs for SUMO. By leveraging
47 the GMNS, a robust framework for standardized network representation (Berg et al., 2022;
48 Lu & Zhou, 2023; Luo, 2024; Luo et al., 2024), utdf2gmns enhances data consistency,
49 reproducibility, and collaboration through four core capabilities: it automates geocoding
50 of Synchro's relative coordinates into accurate longitude–latitude pairs; integrates with the
51 Sigma-X engine (Milan 2022) to extract and optimize key intersection metrics (phasing
52 diagrams, turning volumes, movement capacities, volume-to-capacity ratios, and control delays);
53 generates GMNS-compliant SUMO networks that fully preserve signal coordination, traffic
54 flows, and turning movements; and provides a modular architecture for extension to additional
55 microsimulation platforms, thereby promoting broader standardization and community-driven
56 development.

57 For detailed documentation, please refer to [Official Documentation](#)

58 State of the field

59 Traffic microsimulation scenarios are commonly created from open GIS/map data, built directly
60 in a simulator's native formats, or derived from proprietary planning and operations tools. GIS-
61 based pipelines scale well for network geometry, but key operational inputs (turning movements,
62 lane connectivity, detectors, and signal timing) are typically added manually. When a study
63 starts from Synchro exports, UTDF provides many of these operational details, but they are
64 not available in an open-source simulator-ready representation.

65 SUMO offers flexible signal control modeling and is widely used for research on controller
66 representation and validation (Halbach & Erdmann, 2022; Schrader et al., 2022). These
67 capabilities primarily address how to represent and execute control once a scenario is specified;
68 they do not provide a standard, reusable pathway for translating Synchro's phasing, timing,
69 and coordination plans from UTDF into consistent, reproducible simulation inputs.

70 To our knowledge, there is no commonly used research software package that converts Synchro
71 UTDF directly into a complete, SUMO-ready simulation scenario (network, flows, and signal
72 control). Instead, published and practical workflows are typically assembled from project-
73 specific scripts and manual steps, and they often cover only subsets of the required information
74 (e.g., geometry, turning counts, or a subset of timing). Extending general-purpose GIS/OSM
75 import utilities is therefore insufficient: supporting UTDF requires a dedicated parser and a
76 semantics-preserving mapping of signal control and demand, plus validation tooling to ensure
77 fidelity across intersections and time plans.

78 utdf2gmns provides this missing conversion layer as an end-to-end workflow from Synchro
79 UTDF to a standardized intermediate representation (GMNS) and then to simulation-ready
80 SUMO artifacts. The software's contribution is a reproducible translation of UTDF operational
81 semantics (signal structure, timing/coordination plans, and turning movement demand) into
82 open outputs suitable for large-scale scenario generation and comparative experiments, with
83 optional Sigma-X integration to support intersection-level diagnostics and auditing.

Software design

The `utdf2gmns` package is designed with a modular architecture to facilitate the conversion of Synchro UTDf data into GMNS-compliant networks and SUMO simulations. The core component is the `UTDF2GMNS` class, which orchestrates the entire conversion process through a series of method calls. The software is organized into several key modules:

- **Main Module (`_utdf2gmns.py`):** Contains the primary `UTDF2GMNS` class, which initializes with UTDf file paths and region names. It provides high-level methods such as `geocode_utdf_intersections()`, `utdf_to_gmns()`, and `utdf_to_sumo()` to perform the conversions.
- **Functional Libraries (`func_lib/`):** Divided into submodules for specific functionalities:
 - `utdf/`: Handles reading and processing UTDf data, including geocoding intersections and converting lane data.
 - `gmns/`: Manages the conversion to GMNS format, including node and link generation, and integration with the Sigma-X engine for signal intersection processing.
 - `sumo/`: Generates SUMO-compatible XML files for nodes, edges, connections, and traffic flows, while also handling signal controls and U-turn removal.
 - `plot_net.py`: Provides visualization capabilities using libraries like Matplotlib and Kepler.gl.
- **Utility Libraries (`util_lib/`):** Includes helper functions for tasks such as distance calculations, time conversions, and package settings.
- **Engine (`engine/`):** Integrates the Sigma-X engine for advanced signalized intersection analysis, enabling the extraction of phasing diagrams, turning volumes, and control delays.

This modular design ensures extensibility, allowing users to customize or extend functionalities for other microsimulation platforms. The package leverages external libraries like Pandas for data manipulation and `pyufunc` for utility functions, promoting code reusability and maintainability.

Research impact statement

The `utdf2gmns` package has demonstrated impact in traffic microsimulation by facilitating the conversion of Synchro UTDf data into SUMO-compatible networks. A peer-reviewed Winter Simulation Conference paper presents `utdf2gmns` as an automated workflow for converting UTDf network representation, signalized intersections, and turning volumes into GMNS-compliant SUMO inputs, and reports case-study validation for complex urban corridors (Luo et al., 2025). The same workflow directly addresses the manual Synchro-to-SUMO preparation steps identified in earlier traffic simulation studies (Singh et al., 2017; Udomsilp et al., 2017; Zhang et al., 2024). The subsequent Sim2Signal preprint cites this UTDf-to-SUMO work when describing the construction of its Tempe and Bullhead City benchmark networks from real UTDf signal-plan data (Rafi et al., 2026), providing evidence of scholarly uptake in traffic signal control research.

Furthermore, the integration of `utdf2gmns` with the Sigma-X engine supports visualization and auditing of signalized intersections, including phasing diagrams, turning volumes, volume-to-capacity ratios, and control delays (Luo et al., 2025). These capabilities provide practical tools for urban planners, traffic engineers, and researchers who need reproducible network preparation rather than project-specific scripts.

The DOE-supported Real-Twin project uses `utdf2gmns` for file conversion in its transportation simulation workflow. Real-Twin is supported by the U.S. Department of Energy, Vehicle Technologies Office, Energy Efficient Mobility Systems program under project EEMS114 (Wang et al., 2025). This application provides a concrete example of research use. The package also

131 recorded more than 36,000 PyPI downloads as of June 16, 2026 (PePy.tech, 2026), reflecting
132 distribution through the public package index.

133 The project combines contributions to its conception, implementation, and testing. Xuesong
134 Zhou proposed the initial project concept and supported project funding, while Xiangyong Luo
135 maintains the software implementation. In February 2025, contributor Yiran6 submitted four
136 commits through a public pull request that Luo merged (Yiran6 & xyluo25, 2025). These
137 changes corrected signal-direction mapping, replaced SUMO traffic-light logic directly in the
138 network XML, and added simulation configuration and route files for testing. This development
139 record documents substantive participation by another developer in signal conversion and test
140 preparation.

141 The software has also been refined through independent testing during JOSS review. Public issue
142 reports prompted clearer geocoding dependency errors, complete default runtime dependencies,
143 explicit platform requirements for Sigma-X, and a compatible Setuptools constraint for optional
144 Kepler.gl visualization (anitagraser, 2026; georgiastuart, 2026a, 2026b). These examples show
145 how external feedback has influenced implementation and installation guidance. The repository
146 provides [contribution instructions](#) and [public issue forms](#) for reporting problems, requesting
147 support, and proposing improvements.

148 Hands-On tutorial

```
import utdf2gmns as ug

if __name__ == "__main__":

    region_name = " Bullhead City, AZ"
    path_utdf = "datasets/data_bullhead_seg4/UTDF.csv"

    # Step 1: Initialize the UTDF2GMNS
    net = ug.UTDF2GMNS(utdf_filename=path_utdf, region_name=region_name)

    # Step 2: Geocode intersections from one known intersection coordinate
    single_coord = {
        "INTID": "39",
        "x_coord": -114.59807666698381,
        "y_coord": 35.02605198650903,
    }
    net.geocode_utdf_intersections(single_intersection_coord=single_coord)

    # Step 3: convert UTDF network to GMNS format (csv)
    net.utdf_to_gmns(incl_utdf=True)

    # Step 4: convert UTDF network to SUMO
    net.utdf_to_sumo(sim_name="", remove_U_turn=True, sim_duration=7200)

    # Step 5 (optional): visualize the network
    net_map = ug.plot_net_mpl(net, save_fig=True, fig_name="Bullhead_City.png")
    net_map = ug.plot_net_keplergl(net,
                                    save_fig=True,
                                    fig_name="Bullhead_City.html")

    # Step 6: Sigma-X visualize signalized intersection
```

```
# net.utdf_to_gmns_signal_ints()
```

AI usage disclosure

GitHub Copilot assisted selected repository maintenance changes, including continuous-integration fixes. ChatGPT-5.2 assisted earlier manuscript polishing. The authors are responsible for reviewing the revised text, verifying the reported evidence, and approving the final submission.

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